The Compiler Closure of TFPT

What concretely simplifies, reduces and closes relative to the seven original papers — a status assessment and reading guide

June 11, 2026

In one sentence

The theory tips from "many sectors with surprising hits" (the state of Papers 1–7) to "*one small discrete compiler generates those sectors*": E_8 — which in v4.5 appeared only *peripherally* (the orbit cascade for scale ordering) — is now *built* from two TFPT atoms and moved to the centre; the flavor matrix, the fine-structure constant and the scale grammar follow as *fixed points* from exactly two inputs $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$.

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus Appendix H**, best read in order (this is the entry point; a detailed map is in §1):

1. `introduction` — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

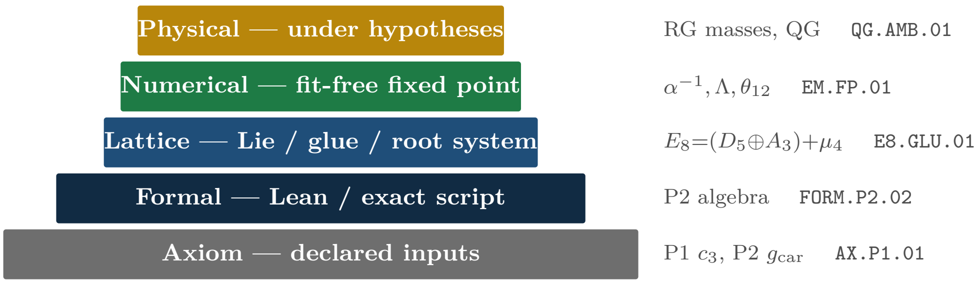
H. `tfpt_horizon_readouts` — **Appendix H** (unit-system reframe, *not* new physics): one seam constant as the universal horizon thermal code.

O. `origin_theory` — **Origin Theory**: why no free fundamental number remains. The $(g_{\text{car}}, N_{\text{fam}})$=$(5, 3)$ skeleton, the triply-forced 8 (geometry = lattice = gravity), the order-30 Coxeter cycle, one boundary transport for flavor and horizon, and the gapped *unique attractor*. [I]/[L] core (`v53`–`v56`) plus one honestly-typed [P] cyclic interpretation.

You are reading document #1.

Contents

Reading the markers (refined). The former overloaded "[M]" is split into the grade it actually carries, so a reviewer can see at a glance which claims are exact, which are formal, and which are conditional: [I] *exact identity* (e.g. $240 = 16 \cdot 5 \cdot 3$); [L] *Lie/lattice theorem* (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$); [F] *formalised* (Lean- or script-verified); [N] *numerical fixed point* (reproduced, stable); [P] *physical/conditional* (follows under named hypotheses); [A] *axiom or open*.



The claim stack (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv` (Fig. 1).

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [I] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L); (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* ("look, another number") into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support)

1. the architecture: the two axioms $\{c_3, g_{\mathrm{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked

(verification/v1_e8_glue.py);

3. the α^{-1} fixed point plus ablation (v3_em_alpha.py);
4. the **single status ledger** verification/status_ledger.csv — the source of truth for every claim's grade, location, dependency and script;
5. the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

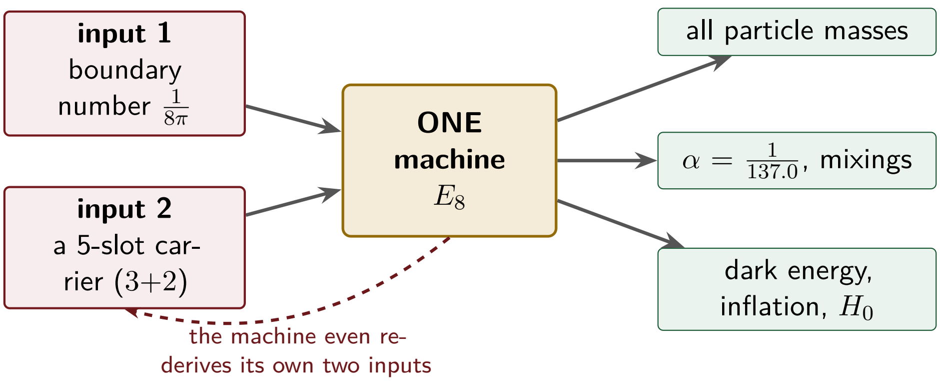
1 What this is about: the jump in construction principle

In plain words — what we found, and why it matters

What we found. The older version of the theory already reproduced dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs. What it could *not* say was *why* the same handful of small integers $(2, 3, 5, 16, \dots)$ kept reappearing in every sector. The new result closes exactly that gap: those numbers are not separate lucky hits. A single tiny "machine" — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8, and E_8 is the unimodular "audit hull" that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not "printed" for free — where a scheme or analytic interface is needed, it is named).

How this drastically simplifies the theory. Instead of seven separate derivations that happen to share numbers, there is now *one generator and a handful of corollaries.* Roughly 250 pages of sector-by-sector work collapse to "two inputs + one machine"; the number of independent structural assumptions drops to **two**.

How plausible it is. A theory becomes more believable when it *explains more from less.* Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed loop below), so there is *no free dial left to tune.* The only genuinely free, continuous ingredient that remains is the number π.

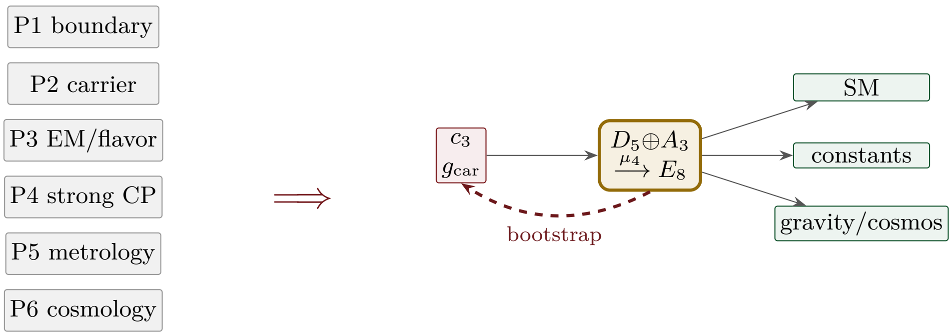


Two numbers go in on the left; the discrete Standard-Model core, the dimensionless constants and several scale readouts come out on the right — and the dashed loop is the point: the machine reproduces the very two numbers it started from, so the discrete core is overdetermined *rather than fitted. (Dimensionful EW/QCD masses, the full quantum-gravity measure and the frontier items stay typed-open — see the status ledger.)*

The seven original papers (tfpt-45, Papers 1–7) deliver a long chain of physical closures: boundary kernel → carrier → Standard-Model packet → EM fixed point → flavor transport →

admissibility/strong CP $\to$ geometric gravity/metrology $\to$ cosmology/CMB. Each stage stands on its own and hits a remarkable number of values. What was missing was a *meta-statement* explaining *why* so many independent sectors use the same small stock of numbers $(2, 3, 5, 16, 40, 41, 48, 240, 248)$.

Version 4.5 — seven papers **New — one compiler**



many parallel sector closures; E_8 only a **peripheral** backdrop

two inputs build E_8; E_8 feeds back and fixes the inputs (closed loop)

The difference in one picture. v4.5 is a *stack of seven independent closures* that happen to share numbers, with E_8 assumed in the background. The new approach is *one discrete compiler*: two inputs build E_8, E_8 outputs the sectors, *and* E_8 feeds back to fix the two inputs (the red loop). What were seven parallel results become one generator with corollaries; what was a posited E_8 becomes a constructed-and-self-checking closure.

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a *compiler hull* [I]/

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\mathrm{car}}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|, \qquad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$ generate the big Lie data directly: $p_1 = 4 = |\mu_4|$, $p_2 = 6 = |R^+(A_3)|$, $p_3 = 10 = \mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \operatorname{rank} E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\mathrm{car}}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$. [`verification/v23_anchor_generator.py`]

And a is itself half-forced ([L]/[P]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\mathrm{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not "why $a = (1, 1, 2)$" but "why the carrier $g_{\mathrm{car}} = 5$ / $(|\mu_4|, N_{\mathrm{fam}}) = (4, 3)$" — that interface (with c_3) is the [A] input layer.

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not "everything in the adjoint". This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which

TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1,1,2)$ into the E_8 audit hull.*

In one line:

$\boxed{\text{The anchor generates the syntax; the boundary generates the scale; } E_8 \text{ checks the consistency.}}$

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1,1,2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0, α^{-1}, the exponential scales) flows from π; and E_8 is the unimodular *checksum container*, not the world group.

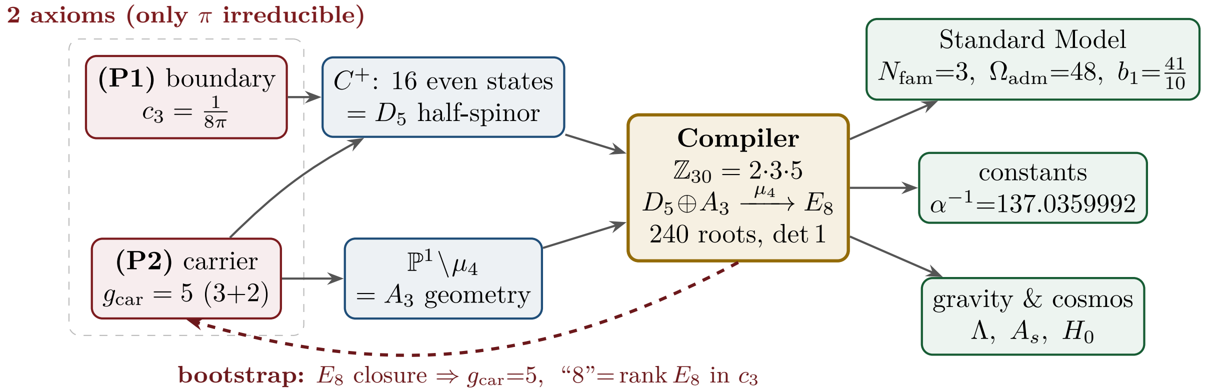
That is now the case. The whole development is consolidated into **five companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the earlier notes of its layer into self-contained parts:

Document	**What it contains (former notes now as parts)**
`tfpt_1_architecture_e8`	**Core.** The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}$=2·3·5, the explicit $D_5 \oplus A_3 + \mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\mathrm{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ (Part II).
`tfpt_2_standard_model`	**Standard Model.** The full SM in one φ_0^{ret}-ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/n_s, r, Starobinsky M, derived θ_{12}, quark c, the H2 splitting type, Λ-typing (Part III).
`tfpt_3_e8_audit_bootstrap`	**E_8 audit & bootstrap.** The seven E_8 slices as an audit raster ($E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade D=60−2n as the same even-integer spine (Part II); the Möbius bootstrap — g_{car}=5 and the "8" in c_3 as overdetermined fixed points, only π irreducible (Part III).
`tfpt_4_frontier`	**Frontier.** The honest status of η_B, m_p/m_e, Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is *not* forced onto the ladder.
`tfpt_horizon_readouts` **(App. H)**	**Appendix H — horizon unit system (reframe, not new physics).** One seam constant c_3=$\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, v_{GW}=c — standard physics in seam units, two compiler fingerprints (1920=$\lvert W(D_5)\rvert$, $\lvert\mu_4\rvert$=4).

The five formerly "research-level" problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\mathrm{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Five companion files (plus this introduction) instead of a dozen, each result where it logically lives.

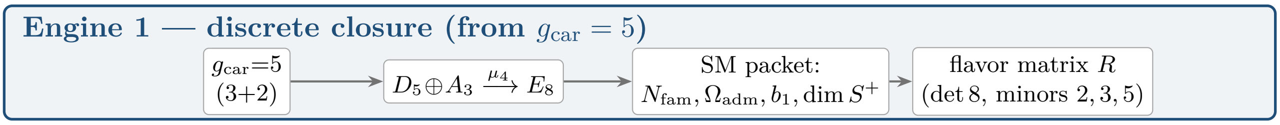
2 The architecture at a glance

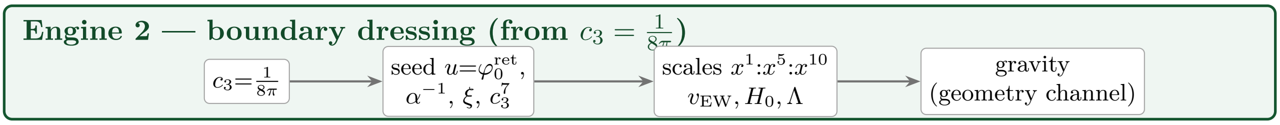
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.



Key statement (the loop closes): previously D_5, A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs*. $g_{\rm car} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0}+\binom{g}{1}+\binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is $\operatorname{rank} E_8 = h(D_5) = \det R$. **Inputs and output are mutually locked — they prove each other**; the only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed self-consistency loop.

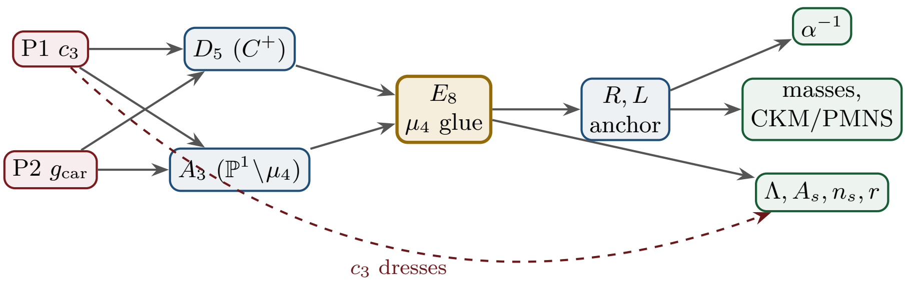
The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.





3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, c_3=$1/(8\pi)$	— (axiom)	[A]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\rm car}$=5	— (axiom)	[A]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[I]	$\dim S^+$=16, Y	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[I]	$N_{\rm fam}$=3, $\mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: disc=$\mathbb{Z}_4$, q-sum= 2	D_5, A_3	[I]	$240, 248$	not even-unimodular
R, L	residue + winding matrix, anchor $(1,1,2)$	A_3	[I]	$\det 8$, minors $(2,3,5)$	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)$=0	c_3, M=41	[N]	137.0359992	no/second root (excl.)
masses	$\varphi_0^{\rm ret}$-ladder $\lambda_Y^L \Lambda$	R, L, c_3	[I]/[P]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam ε=c_3	R, L	[N]/[P]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[P]	Λ, A_s, n_s, r	ambient RP fails

Marker key: ***[I]*** *exact identity ·* ***[L]*** *Lie/lattice theorem ·* ***[F]*** *formalised ·* ***[N]*** *numerical fixed point ·* ***[P]*** *physical/conditional ·* ***[A]*** *axiom / open.*

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

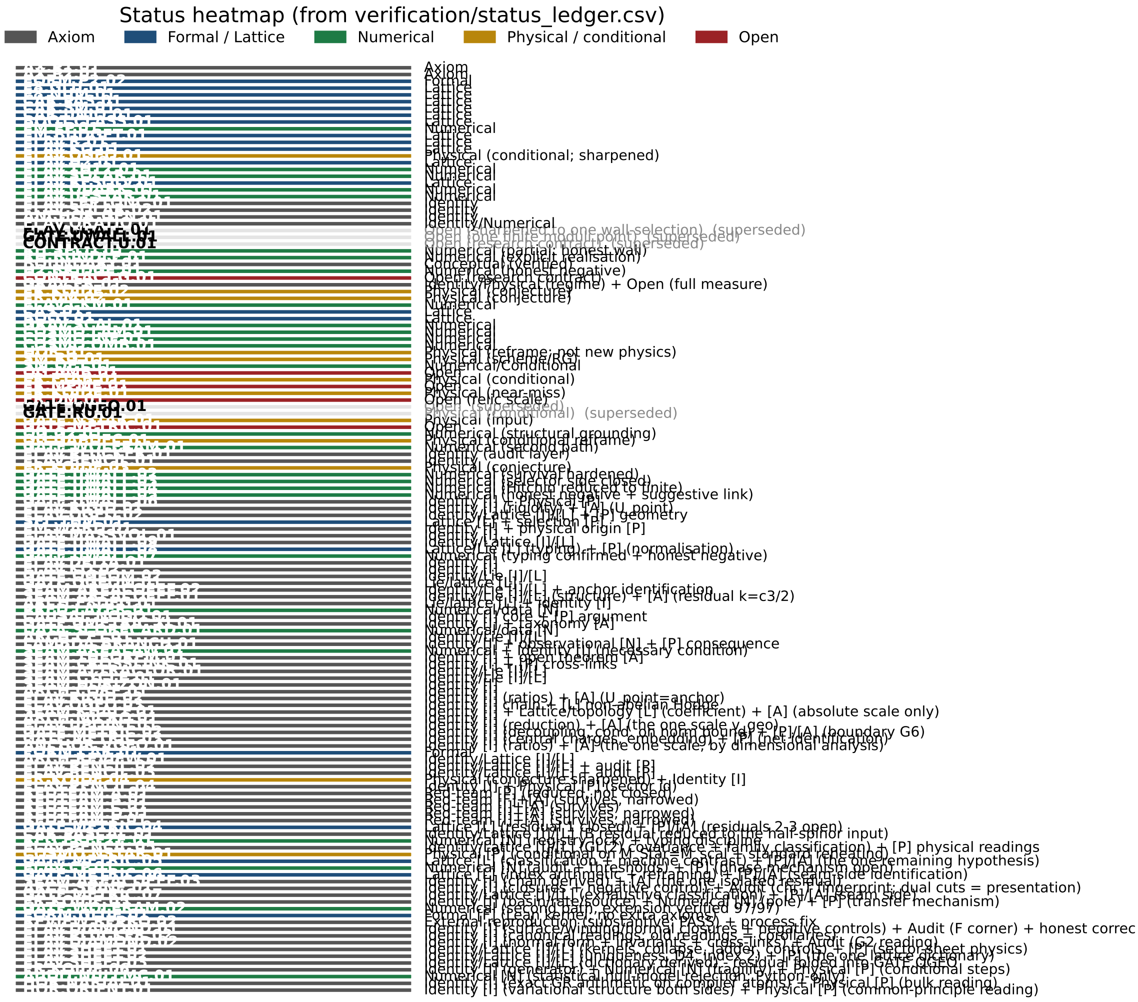


Figure 1: Status heatmap, generated directly from the single source of truth `verification/status_ledger.csv`. Grey = axiom, blue = formal/lattice, green = numerical, gold = physical/conditional, red = open. Each row carries a claim ID (`E8.GLU.01`, ...) that resolves to a ledger row with its dependencies and verification script. The ledger is *versioned*: each row has `active` and `canonical_status`/`supersedes` fields, so claims that were later superseded (e.g. the early "flavor gate open" rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive and drawn dimmed — the canonical status is the only one read as current.

The compact status formula (three honest layers)

$$\boxed{\underbrace{\text{compiler}}_{\text{closed}} \;\Big|\; \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \;\Big|\; \underbrace{\text{strict physical TOE}}_{\text{open}}}$$

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R-bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-*boundary* measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). *(i) Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6 \log \frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([I]/[L] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [P]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). *(ii) Newton's constant is not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* Λ, so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([I]/[P], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just π, the discrete E_8-closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \to v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: everything open is two research gates plus typed interfaces

After the compiler closure, the entire residual is

$$\boxed{\text{Rest} = (U_{\text{wall}}) \;\oplus\; (G_{\text{metric}}) \;\oplus\; (F_{\text{frontier}})}$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity): $(U_{\text{wall}}) \to v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \to G_{\text{net}}$ — reduced to a single boundary-net theorem, *"the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net"* (holomorphy then follows, v89/`GATE.METRIC.06`); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank $= c$: 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [L]; $(F_{\text{frontier}}) \to F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[I]/[L]	gapped transport $\Rightarrow$ *unique* Perron–Frobenius attractor (v56); bootstrap web g_{car}=5 forced (v6/v14)
Newton constant G_N	[I]/[P]	not *independent*: branch pins $\bar{M}_{\text{Pl}}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[I]	$g_{\text{car}}\|\text{Pl}_{\mathbf{1},a}(K)\|_1/(N_{\text{fam}}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
(U) flavor gate	[A]$\rightarrow v_{\text{geo}}$	**Gate 1 closed**: $U_{\text{point}}\rightarrow v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$
$R+R^2$ gravity	[I]/[P]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
physical QG sector	[F]/[P]	gap-decoupled admissible measure (Δ_{eff}=1.648>0)
ambient QG measure	[A](reduced)	**Gate 2 tier B**: holographically reduced to a finite seam-*boundary* measure (v76); IR tier closed (decoupling); strict-TOE cert. open (G_{metric})
Koide	[P]	$Q_{\text{src}}+\varphi_0/24$ as source$\rightarrow$pole transfer conjecture
axion DM	[P]/[A]	$f_a = M_{\text{scal}}/128$ as a scenario
η_B	[P]	readout closed; leptogenesis Boltzmann solve missing
m_p/m_e	[A]	cross-sector QCD/EW ratio
$N_\star$	[P]	reheating input, $N_\star \simeq 57$ (continuous)
$a=(1,1,2)$	[L]/[P]	forced by $4=2+1+1$, conditional on the carrier
carrier selection g=5 (QBL)	[I]/[L] + [P]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank $=c$); residue = the G_{net} premise itself — gate closes $\Rightarrow$ carrier [L]
E_8 Lean	[F]$\rightarrow$[L]	hardening, not a blocker (script-certified)

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Only *two* research gates remain — (U_{wall}) and G_{metric}; the frontier items are typed interfaces, not structural holes. Of these, (U_{wall}) is the only genuine *physics* gate left: G_{metric} is now heat-kernel grounded (G2: the spectral action gives $R+R^2$ structurally) and *gap-decoupled* (G5: Δ_{eff}=1.648>0 isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority (U_{wall}) first. [`verification/v36_spectral_action_g2.py`]

4 Before / after: what concretely changes

Aspect	State of Papers 1–7	After the compiler closure
E_8	*peripheral*: only an $11D\to 4D$ reduction aside and the nilpotent-orbit *cascade* (scale ordering)	**central, constructed**: $C^+{=}D_5$ spinor, $\mathbb{P}^1\backslash\mu_4{=}A_3$, μ_4 glue $\Rightarrow E_8$ [L]
E_8 numbers	$240, 248$ read off tables when needed	$240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$ *derived* as carrier traces [I]
fine structure α^{-1}	*already* a parameter-free cubic fixed point $\alpha^3{-}2c_3\alpha^2{-}8bc_6\ln(\cdot){=}0$, $\alpha^{-1}{\approx}137.0365$	*same* equation, now **existence & uniqueness proved** and refined to 137.0359992; dev. $2.9{\times}10^{-10}$ vs CODATA-2022 (1.9σ) [I]/[N]
flavor residues	partly audit-flagged	anchor $\{1,1,2\}$ & matrix forced by 3 hypotheses [I]/[P]
many sectors	side by side, each its own closure	*one* compiler $\mathbb{Z}_{30}{=}2{\cdot}3{\cdot}5$ generates them [I]
inputs	c_3, g_{car} (+ implicit structure)	exactly 2 axioms (P1, P2), clearly declared [A]
A_s, n_s, r (inflation)	scale present	scalaron mass fixed by c_3^7 ($3.06{\times}10^{13}$ GeV); A_s, n_s, r become *predictions* [P]
θ_{12} (solar)	open (leading $\frac{1}{3}$)	**cond. derived** $\frac{1}{3}-\frac{\varphi_0^{\mathrm{ret}}}{2}=0.307$, given seam $\varepsilon{=}c_3$ [N]/[P]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Reduction in one line

The number of *structural* assumptions drops from "a fixed-point seed c_3, a carrier, the orbit cascade, plus a per-sector calibration in each block" to **two axioms**. Everything else ($N_{\mathrm{fam}}, \Omega_{\mathrm{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) becomes a *consequence*. *And even those two tighten:* the bootstrap note shows $g_{\mathrm{car}} = 5$ and the integer 8 in c_3 are *overdetermined E_8-closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \mathrm{rank}\, E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.)

5 Paper by paper: what each original paper did, and how it works now

The seven original papers (plus the orientation note) total roughly **247 pages**. The table below puts each one next to its compiler-era replacement: *what* it treated and computed, *how many pages* it took, *where* that now lives and how much it compresses, and whether the *accuracy* of the result improves on top. The honest summary up front: in most sectors the *numbers are the same* — what shrinks dramatically is the *derivation length* and the number of independent assumptions; in three places (the solar angle, the inflation amplitude, the scalaron mass) there is a *genuinely new or sharper* result.

P	Treated & computed	Old	How it works now (note; compression)	Acc.
0	orientation: architecture, decoder core $Y, [u_\Sigma], u$, theorem stack	20 p	*one* compiler diagram: $\{c_3, g_{\rm car}\} \to D_5 \oplus A_3 \to E_8$ (intro + action_grammar)	=
1	boundary kernel: $c_3 = \frac{1}{8\pi}$, Calderón polarization, carrier decoder $6Y^2 - Y - \mathbf{1} = 0$	20 p	the two axioms P1 (c_3) + P2 ($g_{\rm car}$=5); carrier = 3+2 atom (compiler)	=
2	carrier/SM packet: S^+=16, $G_{\rm phys}$, $N_{\rm fam}$=3, $\Omega_{\rm adm}$=48, hypercharge, b_1	48 p	compiler reads 16=1+5+10, $N_{\rm fam}$, $\Omega_{\rm adm}$, $b_1 = \frac{41}{10}$ as carrier traces; $C^+ = D_5$ spinor (∼5 p)	=
3	EM closure: α^{-1} via $\sum L + N_\Phi = 41$; flavor transport, CKM/PMNS, θ_{13}	38 p	α^{-1} as unique root of $F_{U(1)}(\alpha)$=0; flavor from transport theorem (parabolic + sm)	↑
4	admissibility/strong CP/QFT: $\theta_{\rm eff}$=0, OS reconstruction, mass gap	40 p	explicit gap $6 \log \frac{3}{2} \Rightarrow$ Dobrushin/OS; strong CP from structure+RP (closing_comp.)	=
5	geometric Hodge/metrology: $M_{\rm Pl}$, $v_{\rm geo}$, Λ, spectral Einstein	30 p	Hodge/R^2 branch; Λ *typed* (one branch label); boundary-normalised (closing_comp.)	=
6	cosmology interface: FRW, $\Lambda_{\rm IR}$, scalaron, reheating, leptogenesis, axion	15 p	Starobinsky $M = c_3^{7/2} \bar{M}_{\rm Pl} = 3.06 \times 10^{13}$ GeV; A_s *parameter-free* (closing_comp.)	↑new
7	CMB operational closure: C_ℓ, $a_{\ell m}$, seam mixer, Planck pipeline	36 p	n_s=0.964, r=0.0040 from the R^2 attractor, no free amplitude (closing_comp.)	↑new

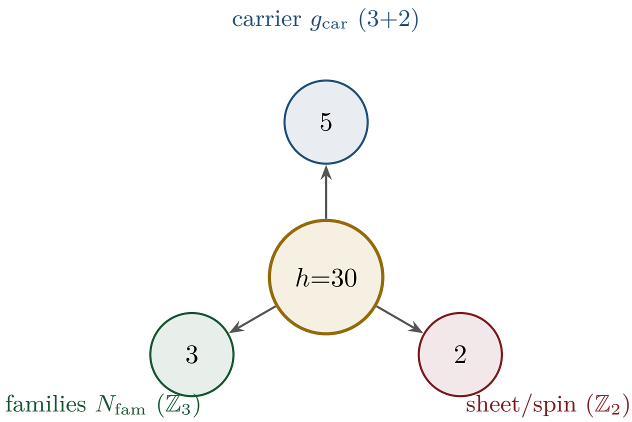
Three places where accuracy genuinely improves "on top"

1. **Solar angle θ_{12}.** Paper 3 left it as the only open SM number (leading $\frac{1}{3}$). It is now *conditionally derived* (modulo the seam-misalignment lemma): TBM gives $\frac{1}{3}$, the charged-lepton 1–2 misalignment is the seam $\varepsilon = \frac{N_{\rm fam}}{|\mu_4|} \varphi_0^{\rm ret} = \frac{3}{4} \varphi_0^{\rm ret}$ (leading term exactly $c_3 = \frac{1}{8\pi}$), and TBM geometry gives $\sin^2 \theta_{12} = \frac{1}{3} - \frac{2}{3}\varepsilon = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.3067$ — 0.1% from NuFIT.
2. **Inflation amplitude A_s.** Generic Starobinsky fits the scalaron mass to A_s. TFPT fixes it by the seam, $(M/\bar{M}_{\rm Pl})^2 = c_3^7$, so $A_s = \frac{N_\star^2}{24\pi^2} c_3^7 = 2.0 \times 10^{-9}$ is a *prediction* (Planck 2.1×10^{-9}), and the measured A_s even *predicts* $N_\star = 56$.
3. **Scalaron mass.** $M = c_3^{7/2} \bar{M}_{\rm Pl} = 3.06 \times 10^{13}$ GeV comes out *exactly* at the canonical Starobinsky value — a number that was an input before is now an output.

What "simpler" means concretely. The compression is not cosmetic. Paper 2 spent 48 pages establishing the carrier/SM packet by rigidity arguments; the compiler reads the same 16=1+5+10, $N_{\rm fam}$=3, $\Omega_{\rm adm}$=48, $b_1 = \frac{41}{10}$ off the Pascal row of a 5-slot carrier in a page. Paper 3's flavor transport (38 p) becomes one transport theorem plus a fixed residue matrix whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers. The "many independent closures" of Papers 1–7 become *one* generator with a handful of corollaries.

6 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

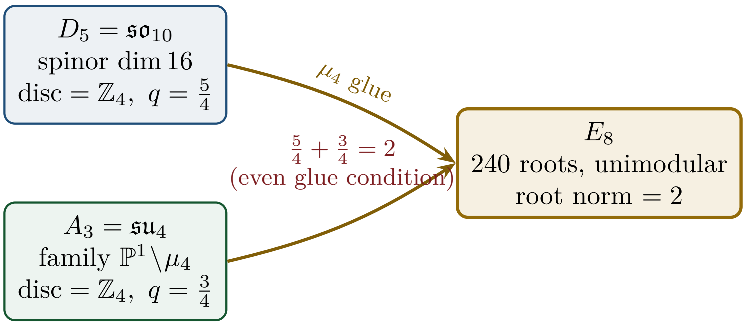


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. [I]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. [I]
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30}; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \backslash \mu_4$ is the A_3 Coxeter element. [I]

7 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [L]

8 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\rm car}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, $\dim S^{+}{=}16$, $N_{\mathrm{fam}}{=}3$, $\Omega_{\mathrm{adm}}{=}48$	[I]
hypercharge	roots of the charge polynomial $bsY^2 + (s{-}b)Y - 1$	[I]
$U(1)$ index b_1	$b_1 = \dfrac{g_{\mathrm{car}}\, 2^{g_{\mathrm{car}}-2} + 1}{10} = \dfrac{41}{10}$	[I]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[I]/[N]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[I]/[P]
electroweak scale	$v_{\mathrm{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[I]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[I]/[P]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[P]
inflation amplitude	$A_s = \dfrac{N_\star^2}{24\pi^2} c_3^7$ in the R^2 branch	[P]
E_8	$D_5 \oplus A_3 + \mu_4$-glue	[L]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by $5 =$ carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

Typing note on Λ (a deliberate danger zone). "Λ" denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [I] readout), (ii) the dimensionless density quotient $\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4$ (which carries a prefactor *branch*, hence [P]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier — not a separate module

With $x = v_{\mathrm{geo}}/(5\beta_{\mathrm{rad}}^2 \bar{M}_{\mathrm{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5:

$$\boxed{\frac{H_0}{\bar{M}_{\mathrm{Pl}}} = x^5 R_H, \qquad \frac{\rho_\Lambda}{\bar{M}_{\mathrm{Pl}}^4} = x^{10} R_\Lambda}, \qquad (1,5,10) = \tbinom{5}{0}, \tbinom{5}{1}, \tbinom{5}{2}.$$

EW is the unit x^1, Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. "Hubble = slot product, Λ = pair product." [I]

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\mathrm{geo}}}{\sqrt{2}}\, \lambda_Y^{L_{f,j}}\, \Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}, \qquad \varphi_0^{\mathrm{ret}} = \tfrac{1}{6\pi} + \tfrac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers ($\mathrm{tr} = 9 = N_{\mathrm{fam}}^2$, $\det = 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the φ_0^{ret}-ladder $\hat{m} = \frac{v_{\mathrm{geo}}\pi}{\sqrt{2}} c\,(\varphi_0^{\mathrm{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the (U_{wall}) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2}$; see below). *(Full derivation: document 2,* `tfpt_2_standard_model`*.)*

9 Physical, geometric, topological: which solution is more plausible?

Level	**E_8 peripheral (v4.5)**	**E_8 compiled (new)**
physical	α^{-1} already a cubic fixed point, but each sector closed on its own	α^{-1} as the fixed point of *one* equation; one generator for many scales
geometric	Lie algebra external; E_8 only orders the cascade	the C^+ spinor + puncture geometry $\mathbb{P}^1 \setminus \mu_4$ give D_5, A_3, glued to E_8
topological	—	the common discriminant $\mathbb{Z}_4$ (μ_4 winding) forces the glue; Φ_{30}/Coxeter drives the sectors
parsimony	many structures	2 axioms, the rest a consequence

Why the compiled solution is more plausible

It is (i) *more parsimonious* (fewer independent assumptions ⇒ less overfitting risk), (ii) *more rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *more checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

10 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers $240, 248$ are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain "everything"; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8.

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit:

$E_6 \times A_2$ reads the flavor matrix

$$248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\,a)\,N_{\mathrm{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

11 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R, the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\mathrm{Spec}(Q_+)$ from D_4, v69/v70), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point}, an anchor; Research Contract 1). [I]/[A]
2. **Gravity — exact in the R^2 regime.** Spectral action $\to R+R^2$, BRST/Hodge kills the gauge directions, FRW reduction $\to$ Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n\geq 3} \to R^2$, valid for curvature $\ll M^2$). [I] in the regime
3. **Strong CP — decoupled.** $\theta_{\text{eff}}=0$ follows from three structural facts (γ_5-Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [I] (given RP)
4. **Full dynamics — the admissible transfer model is closed.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector; the continuum and metric-coupled theory still require the stated RP, cluster and limit assumptions. [I] (transfer model) / [P] (continuum)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (`1886 jobs`, 0 `sorry`, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [I]/[A]

In other words: the theory has shrunk from "many open audit points" to "*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms". The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. *(Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.)*

The TOE-completion ledger — what is done, and what genuinely remains

A theory-of-everything must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	**Status**	**If not complete: what closes it**
gauge group, N_{fam}, hypercharge, b_1	[I]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[L]	— complete
bootstrap $(g, \mu) = (5, 4)$, "8" in c_3	[L]	— complete (reverse glue $\mu^2 - 5\mu + 4 = 0$)
fine structure α^{-1}	[N]	— **existence+uniqueness proved**; value is a numerical fixed point (ledger `EM.FP.01`), Ward origin [P]
fermion masses (φ_0^{ret}-ladder)	[I]/[A]	leptons exact; quark *ratios* closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$	[N]/[P]	θ_{12} cond. derived; θ_{23} maximal
flavour matrix / H2	[I]	up to Mehta–Seshadri unitarity (U), a Paper-3 input
inflation A_s, n_s, r, scalaron M	[N]	parameter-free given $M_{\text{Star}} = M_{\text{scal}}$
genuinely open — the actual TOE remainder		
dim. EW/QCD masses	[P]	standard RG threshold matching (no new physics)
QFT measure (admissible sector)	[I]	**closed: RP from cusp spectrum, OS reconstruction**
ambient QFT + full gravity (metric sector)	[A]	*not constructed here* (frontier doc); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	[P]	**hardened:** $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K) = 1/(8\pi)$ **(Gauss–Bonnet)**; "8" = rank E_8; π stays the one primitive
η_B, m_p/m_e, Koide, dark matter	[A]	frontier handles only (see document 4)

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret}-ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed); the *ambient/metric-sector* measure — i.e. *full quantum gravity* — is *not* constructed here and stays [A] (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*: $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K\, dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete "8" equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and π is the lone continuous primitive.

12 Does this make the theory more probable?

Plausibility balance

Pro: (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors $\Rightarrow$ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}$, modulo the seam-misalignment lemma). A final verdict "it maps reality" requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

13 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	TFPT value & derivation	Experiment (running / upcoming)	St.
solar angle $\sin^2\theta_{12}$	$\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}=0.3067$; TBM + seam misalignment $\varepsilon=\frac{3}{4}\varphi_0^{\rm ret}\approx c_3$ (cond.)	**JUNO** (taking data since Aug 2025; sub-percent θ_{12}) — the sharpest live test	[N]/[P]
reactor angle $\sin^2\theta_{13}$	$\varphi_0^{\rm ret}e^{-5/6}=0.0231$; seed × carrier-trace $e^{-\gamma}$, $\gamma=\frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[I]
atmospheric $\sin^2\theta_{23}$	$\approx\frac{1}{2}$ ($\mu\tau$-symmetric limit; octant not selected)	NOvA / T2K / **DUNE** (octant ambiguity, NuFIT 6.0)	[P]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)$=0 (no fit, no second branch)	CODATA-2022 137.035999177(21): dev. 2.9×10^{-10} from the recommended adjustment (1.9σ of its uncertainty)	[I]/[N]
tensor ratio r	$r=\frac{12}{N_\star^2}=0.0040$ ($N_\star$=55); R^2 attractor, $M=c_3^{7/2}\bar{M}_{\rm Pl}$	now $r<0.036$ (BICEP/Keck BK18); **CMB-S4** ($\sigma_r\le5\times10^{-4}$, obs. $\sim$2033)	[P]
tilt n_s	$n_s=1-\frac{2}{N_\star}=0.964$ (same attractor)	Planck (0.965); CMB-S4 sharpens	[P]
amplitude A_s	$\frac{N_\star^2}{24\pi^2}c_3^7=2.0\times10^{-9}$, parameter-free	Planck (2.1×10^{-9})	[P]
neutron EDM	$\theta_{\rm eff}=0\Rightarrow d_n$ essentially null (structure + RP)	**PSI nEDM** / SNS (running)	[I]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	**LEGEND / nEXO**, DUNE, KATRIN	[P]
flavor invariants	$\det R=h(D_5)$=8, principal minors $(2,3,5)$, $\chi_R=t^3-9t^2+10t-8$ (exact)	any future CKM/PMNS global fit must satisfy these	[I]
H_0 vs. Λ	$v_{\rm EW}\sim e^{-\alpha^{-1}/5}$, $\Lambda\sim e^{-2\alpha^{-1}}$, $H_0\sim\sqrt{\Lambda}$: one engine	SH0ES / **DESI** / Planck (Hubble tension)	[P]

*Marker key: **[I]** exact identity · **[L]** Lie/lattice theorem · **[F]** formalised · **[N]** numerical fixed point · **[P]** physical/conditional · **[A]** axiom / open.*

The two decisive near-term tests

(1) JUNO and θ_{12}. JUNO is *taking data since August 2025* and targets sub-percent $\sin^2\theta_{12}$. TFPT commits to the conditional $\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}=0.3067$ (seam $\varepsilon=c_3$). A measured central value away from ~0.307 at high significance falsifies the seam mechanism — a *live* test.

(2) r. The R^2 scalaron with M fixed by c_3^7 predicts $r=0.0040$ ($N_\star$=55), already below the current bound $r<0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r\le5\times10^{-4}$, first observations $\sim$2033–34). A detection of $r\gtrsim0.01$ would be incompatible with the R^2 branch on which $M_{\rm Pl}$ and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two

TFPT atoms really generate E_8; and the discipline rule "*every load-bearing number appears in at least one E_8 projection*" (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge itself is additionally *quantified* by an explicit null model: exhaustively enumerating a declared formula grammar (provably containing every scored TFPT formula) against conservative data windows gives a joint look-elsewhere-corrected null probability $\leq 10^{-30.7}$ that a random theory of equal formula complexity reproduces the scorecard — a statistical statement conditional on the declared grammar, not a claim of certainty [N] [`verification/v100_numerology_null_mc.py`].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying $M_{\rm Pl}, A_s$;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\rm eff} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula↔value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor ($U_{\rm point}$) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+;
- a sharper geometry↔gravity bridge once the Hodge/R^2 branch is lifted from [P] to [I].

Conclusion

The state of Papers 1–7 was an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\rm car}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [I] (exact identity), [L] (Lie/lattice theorem) or [N] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map.

Script	What it machine-checks
`v1_e8_glue.py`	glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$
`v2_carrier_pascal.py`	$g_{\mathrm{car}}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\mathrm{fam}}{=}3$, rank $E_8{=}8$, $\Omega_{\mathrm{adm}}{=}48$, $b_1{=}\frac{41}{10}$
`v3_em_alpha.py`	EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation
`v4_flavor_matrix.py`	residue matrix R: det $=8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF diag$(1,1,8)$, $\lVert R\rVert_F^2{=}78$, $\sum L{=}40$
`v5_e8_cascade.py`	cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to 240$, IR tail 46080, variance 6552
`v6_bootstrap.py`	reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\mathrm{car}}{=}5$ three ways; $8{=}$rank $E_8{=}h(D_5){=}\varphi(30)$
`v7_gravity_cosmo.py`	scalaron exponent 7, $M^2_{\mathrm{scal}}/\bar{M}^2_{\mathrm{Pl}}{=}c_3^7$, A_s, n_s, r, Ω_b, $\sin^2\theta_{12}$
`v8_horizon.py`	horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, S_{dS}, Page, $\beta_{\mathrm{rad}}{=}0.2424°$
`v9_neutrino_texture.py`	solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$, $\theta_{23}{=}45°$, $\theta_{13}{=}0$
`v10_projection_ involution.py`	Q,Σ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; χ_Q, χ_K; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top Ka{=}41$
`v11_unique_KQ.py`	K, Q are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
`v12_mass_ generation_ polynomials.py`	sector/generation polynomials of K and the anchor-block det ladder $(9, 10, 16, 40)$
`v13_open_gates.py`	gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\mathrm{fam}}$
`v14_carrier_ uniqueness.py`	$g_{\mathrm{car}}{=}5$ unique ($N_{\mathrm{fam}} \in \mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\mathrm{Tr}\,Y{=}\,\mathrm{Tr}\,Y^3{=}0$
`v15_bootstrap_ classification.py`	$D_5{\oplus}A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
`v16_solar_dual_ anchor.py`	$a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$; full PMNS
`v17_hexagonal_ resolvent.py`	finite resolvent $D_y^{-1}{=}(\sum y^{5-m}\delta^m U_6^m)/(y^6{-}\delta^6)$ (quark c backbone)
`v18_quark_yukawa.py`	quark source ratios $\frac{m_u}{m_d}{=}0.470085$, $\frac{m_c}{m_s}{=}13.61$, $\frac{m_t}{m_b}{=}40.80$ exact; lepton $c{=}(\frac{16}{7},\frac{4}{3},\frac{7}{6})$; full hierarchy
`v19_monodromy_ moduli.py`	exact pole $z_\star{=}\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4-fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
`v20_lepton_c_ derivation.py`	lepton c's derived: $\delta{=}\frac{1}{2}$ resolvent $c{=}\lvert\mu_4\rvert^w/(\frac{5}{4}{-}\cos\frac{r\pi}{3}) \Rightarrow \frac{16}{7}, \frac{4}{3}$; product $\frac{2^{g_{\mathrm{car}}}}{N^2_{\mathrm{fam}}}{=}\frac{32}{9} \Rightarrow \frac{7}{6}$
`v21_solar_product_ quark.py`	solar coeff $= q(A_3){=}\frac{3}{4}$ (glue-norm, $q(D_5){+}q(A_3){=}2$), $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$; product $\frac{2^{g_{\mathrm{car}}}}{N^2_{\mathrm{fam}}}$; quark law fails ($\frac{1}{7}$ vs 0.724)
`v22_open_gates_ audit.py`	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
`v23_anchor_ generator.py`	anchor $a{=}(1,1,2)$: $e_k{=}(4,5,2)$, $c_3{=}\frac{1}{2e_1\pi}$; $p_n{=}2{+}2^n \Rightarrow 240, 248$, binary ladder; inputs $\to \{a,\pi\}$
`v24_quark_ratio_ closure.py`	quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational c gauge, Λ_q in $O(1)$
`v25_frontier_ conjectures.py`	conjectures: Koide $Q_{\mathrm{src}}{+}\frac{\varphi_0^{\mathrm{ret}}}{24}{\approx}\frac{2}{3}$; axion $f_a{=}M_{\mathrm{scal}}/128{\to}m_a{\approx}23.8\,\mu$eV
`v26_ flavor_frontier_ unification.py`	the "11" is not uniquely forced ($\Rightarrow$ [P]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
`v27_wall_ representative.py`	explicit balanced wall rep W_{wall} (cols = perm$(0,1,2)$, rows $w{=}(2,1,1)$, pardeg 0); selector det $R{=}8$, Spec$(Q_+){=}\{1,2,3\}$
`v28_gravity_fR.py`	$R{+}R^2$ closed: $f(R)$, $M{=}c_3^{7/2}\bar{M}_{\mathrm{Pl}}$ scalaron, $N_\star{=}57{\to}n_s, r, A_s$; full metric measure open; $248c_3^2{<}6\log\frac{3}{2}$
`v29_ research_contract_`	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296$\to$144$\to$5 orbits; symmetry alone does not pick W_{wall}) and $G5$ gap-dominance $2\lVert V\rVert{<}\Delta$

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim↔script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all six documents are written against). The [**F**] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).